Assignment:-

Innorative Method Library Research

Page No. (1)

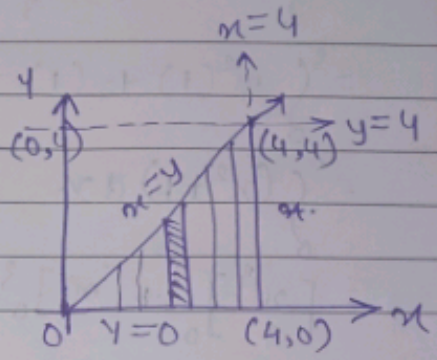
Name:- Rahul K. Ekhar
subject:- C & V
Roll No:- 51
section:- I

Q.1] Evaluate by changing order of integration

$$\int_0^4 \int_y^4 \frac{x}{x^2+y^2}\, dx\, dy$$

Soln:-

$$\int_{y=0}^{y=4} \int_{x=y}^{x=4} \frac{x}{x^2+y^2}\, dx\, dy$$



We have,

$x = 4$, $x = y$

$y = 4$, $y = 0$

$$\int_0^4 \int_y^4 \frac{x}{x^2+y^2}\, dx\, dy = \int_0^4 \int_0^x \frac{x}{x^2+y^2}\, dx\, dy$$

$$= \int_0^4 x \int_0^x \frac{1}{x^2+y^2}\, dy\, dx$$

$$= \int_0^4 x \left[\frac{1}{x} \tan^{-1}\left(\frac{y}{x}\right)\right]_0^x dx$$

$$= \int_0^4 \frac{\pi}{4}\, dx$$

$$= \frac{\pi}{4}(x)_0^4 = \frac{\pi}{4} \times 4$$

$$\int_0^4 \int_y^4 \frac{x}{x^2+y^2}\, dx\, dy = \pi$$

Shot on Y30
AI Quad Camera
2024.12.13 21:34

Q.2] Evaluate $\int_0^3 \int_0^1 (x^2+3y^2)\,dy\,dx$

Soln:- Let,

$$I = \int_0^3 \int_0^1 (x^2+3y^2)\,dy\,dx$$

$$= \int_0^3 \int_{y=0}^{y=1} (x^2+3y^2)\,dy\,dx$$

$$= \int_0^3 x^2[y]_0^1 + \frac{3}{3}[y^3]_0^1\,dx$$

$$= \int_0^3 x^2(1) + (1)\,dx$$

$$= \int_0^3 (x^2+1)\,dx$$

$$= \left[\frac{x^3}{3}\right]_0^3 + [x]_0^3$$

$$= \frac{27}{3} + 3$$

$$I = 12$$

$$\therefore \int_0^3 \int_0^1 (x^2+3y^2)\,dy\,dx = 12.$$

Q.3] Evaluate $\int_0^1 \int_0^y xye^{-x^2}\,dy\,dx$

Soln:-

Let, $I = \int_0^1 \int_0^y xye^{-x^2}\,dy\,dx$

$$= \int_0^1 \int_{x=0}^{x=y} xye^{-x^2}\,dx\,dy$$

$$= \int_0^1 \frac{-y}{2}\left[\int_0^y -2xe^{-x^2}\,dx\right]dy$$

$$= \frac{1}{2}\int_0^1 -y\left[e^{-x^2}\right]_0^y dy$$

$$= \frac{1}{2}\int_0^1 -y\left[e^{-y^2}-1\right] dy$$

$$= \frac{1}{2}\int_0^1 \left[y - ye^{-y}\right] dy$$

$$= \frac{1}{2}\left[\frac{y^2}{2} + \frac{1}{2}e^{-y^2}\right]_0^1$$

$$= \frac{1}{4}\left[y^2 + e^{-y^2}\right]_0^1$$

$$= \frac{1}{4}\left[1 + \frac{1}{e} - 0 - 1\right]$$

$$= \frac{-1}{4e}.$$

$$\therefore \int_0^1\int_0^y xy\, e^{-x^2} dy\, dx = \frac{-1}{4e}.$$

4] evaluate $\int_1^2\int_0^1\int_{-1}^1 (x^2+y^2+z^2)\, dx\, dy\, dz$

Sol:- $\int_{z=1}^{z=2}\int_{y=0}^{y=1}\int_{x=-1}^{x=1} (x^2+y^2+z^2)\, dx\, dy\, dz$

$$= \int_1^2\int_0^1 \left[\frac{x^3}{3} + (y^2+z^2)x\right]_{-1}^1 dy\, dz$$

$$= \int_1^2\int_0^1 \left[\frac{1}{3} + (y^2+z^2) - \left(\frac{-1}{3}\right) + (y^2+z^2)(-1)\right] dy\, dz$$

$$= \int_1^2 \left[\int_0^1 \frac{2}{3} + 2(y^2+z^2)\, dy\right] dz$$

$$= \int_1^2 \frac{2}{3}[y]_0^1 + 2\left[\left(\frac{y^3}{3}\right)_0^1 + z^2[y]_0^1\right] dz$$

$$= \int_1^2 \frac{4}{3} + 2z^2 \, dz$$

$$= \frac{4}{3}[z]_1^2 + \frac{2}{3}[z^3]_1^2$$

$$= \frac{4}{3}(2-1) + \frac{2}{3}[8-1]$$

$$= \frac{8}{3} - \frac{4}{3} + \frac{16}{3} - \frac{2}{3}$$

$$= \frac{18}{3}$$

$$= 6$$

Q.5] Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x \, dz \, dx \, dy$

Soln:- $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x \, dz \, dx \, dy$

$$= \int_0^1 \int_{y^2}^1 x[z]_0^{1-x} \, dx \, dy$$

$$= \int_0^1 \int_{y^2}^1 x[1-x] \, dx \, dy$$

$$= \int_0^1 \int_{y^2}^1 (x - x^2) \, dx \, dy$$

$$= \int_0^1 \left[\int_{y^2}^1 x \, dx - \int_{y^2}^1 x^2 \, dx \right] dy$$

$$= \int_0^1 \left[\frac{1}{2}[x^2]_{y^2}^1 - \frac{1}{3}[x^3]_{y^2}^1 \right] dy$$

$$= \int_0^1 \frac{1}{2}(1 - y^4) - \frac{1}{3}[1 - y^6] \, dy$$

$$= \int_0^1 \left(\frac{1}{2} - \frac{y^4}{2} - \frac{1}{3} + \frac{y^6}{3}\right) dy$$

$$= \int_0^1 \frac{1}{2} - \frac{1}{3} - \frac{y^4}{2} + \frac{y^6}{3}\, dy$$

$$= \int_0^1 \frac{1}{6} - \frac{y^4}{2} + \frac{y^6}{3}\, dy$$

$$= \frac{1}{6}[y]_0^1 - \frac{1}{10}[y^5]_0^1 + \frac{1}{21}[y^7]_0^1$$

$$= \frac{1}{6} - \frac{1}{10} + \frac{1}{21}$$

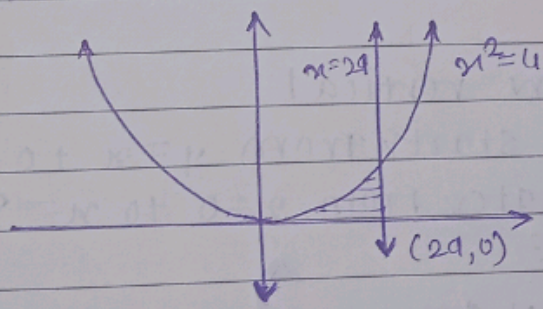
$$= \frac{35 - 21 + 10}{210}$$

$$= \frac{4}{35}$$

Q.6 Evaluate $\iint_R xy\, dx\, dy$, where R is the region bound by axis, coordinate $x = 2a$ and curve $x^2 = 4ay$

Soln:-

Given that region R is bounded by curve $x^2 = 4ay$ & $x = 2a$ as shown in following.



$x^2 = 4ay$

$(2a)^2 = 4ay$

$\boxed{y = a}$

limits are, $x = 2\sqrt{ay}$ to $x = 2a$

$y = 0$ to $y = a$

$$I = \int_{y=0}^{a} \left[\int_{2\sqrt{ay}}^{2a} xy\,dx \right] dy$$

$$= \int_0^a \left(\frac{y}{2} [x^2]_{2\sqrt{ay}}^{2a} \right) dy$$

$$= \int_0^a \frac{y}{2} \left[(2a)^2 - (2\sqrt{ay})^2 \right] dy$$

$$= \int_0^a \frac{y}{2} (4a^2 - 4ay)\,dy$$

$$= \int_0^a (2a^2 y - 2ay^2)\,dy$$

$$= \frac{2}{2} a^2 [y^2]_0^a - \frac{2a}{3} [y^3]_0^a$$

$$= a^2(a^2) - \frac{2a}{3}[a^3]$$

$$= a^4 - \frac{2}{3} a^4$$

$$= \frac{a^4}{3}$$

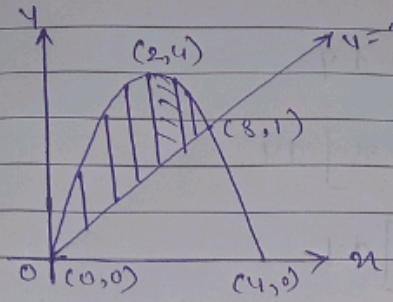
Q.7] find the area lying between the area $y = 4x - x^2$ and the line $x = y$.

soln:

Here, we draw vertical strip which starts from $y = x$ to $y = 4x - x^2$

Again it rotates from $x = 0$ to $x = 3$

$$\therefore \int_{x=0}^{x=3} \int_{y=x}^{y=4x-x^2} dy\,dx$$



for $x=1$ $(1,3)$
$y=4-1, \; y=3$
for $x=2$ $(2,0)$
$y=8-4=4$
for $x=4$ $(4,0)$
$y=0$
for $x=0$ $(0,0)$
$y=0$

$$\therefore \int_0^3 [y]_x^{4x-x^2} \, dx$$

$$= 3\int_0^3 4x - x^2 - x \, dx$$

$$= 3\int_0^3 3x - x^2 \, dx$$

$$= \frac{3}{2}[x^2]_0^3 - \frac{1}{3}[x^3]_0^3$$

$$= \frac{3}{2}[9] - \frac{1}{3}[27]$$

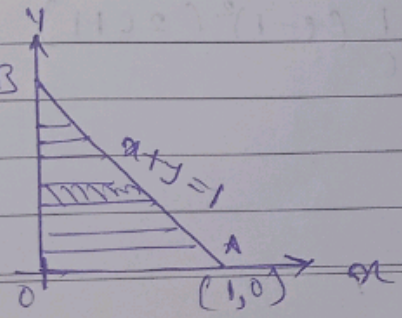
$$= \frac{27}{2} - \frac{27}{3}$$

$$= \frac{9}{2}$$

Q.8] Evaluate $\iint e^{2x+3y} \, dx \, dy$, over the triangle bound by $x=0$, $y=0$ and $x+y=1$

Soln: $\iint e^{2x+3y} \, dx \, dy$

$\therefore x+y=1$
$\therefore$ for $x=1$
$y=1$
$\therefore A(1,0)$, $B(0,1)$



$$\therefore \int_{y=0}^{y=1} \int_{x=0}^{x=1-y} e^{2x+3y}\, dx\, dy$$

$$= \int_0^1 \left[\int_0^{1-y} e^{2x} e^{3y}\, dx \right] dy$$

$$= \int_0^1 \left[e^{3y} \int_0^{1-y} e^{2x}\, dx \right] dy$$

$$= \int_0^1 e^{3y} \left[\frac{e^{2x}}{2} \right]_0^{1-y} dy$$

$$= \int_0^1 \frac{e^{3y}}{2} \left[e^{2(1-y)} - e^0 \right] dy$$

$$= \int_0^1 \frac{e^{3y}}{2} \left[e^{2-2y} - 1 \right] dy$$

$$= \frac{1}{2} \int_0^1 e^{y+2} - e^{3y}\, dy$$

$$= \frac{1}{2} \left[e^{y+2} - \frac{e^{3y}}{3} \right]_0^1$$

$$= \frac{1}{2} \left[e^3 - \frac{e^3}{3} - e^2 + \frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{2e^3}{3} - e^2 + \frac{1}{3} \right]$$

$$= \frac{1}{6} \left[2e^3 - 3e^2 + 1 \right]$$

$$= \frac{1}{6} (e-1)(2e^2 - e - 1)$$

$$= \frac{1}{6} (e-1)^2 (2e+1)$$

Q.3 $\iint \frac{xy}{\sqrt{1-y^2}}\,dx\,dy$ over the positive quadrant of circle $x^2+y^2=1$

Sol^n:- put, $\sqrt{1-y^2}=t$

$1-y^2=t^2$

$-2y\,dy=2t\,dt$

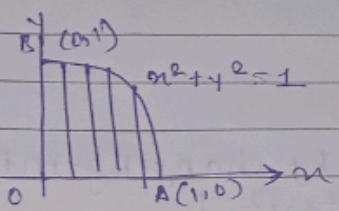
$y\,dy=-t\,dt$

$x^2+y^2=1$

$\therefore y=1, x=1$

A(1,0)

B(0,1)



$$\int_{x=0}^{x=1}\left[\int_{y=0}^{y=\sqrt{1-x^2}}\frac{xy}{\sqrt{1-y^2}}\,dy\right]dx$$

consider, $I=\int_0^{\sqrt{1-x^2}}\frac{xy}{\sqrt{1-y^2}}\,dy$

put, $1-y^2=z^2$

$-2y\,dy=2z\,dz$

$dy=\frac{-z}{y}\,dz$, where,

$y=0 \longrightarrow t=1$

$y=\sqrt{1-x^2} \longrightarrow t=x^2$

$\therefore \int_1^{x^2}\frac{xy}{z^2}\cdot\frac{z}{y}\,dz$

$\therefore x\int_1^{x^2}-1\,dz$

$\therefore -x[z]_1^{x^2}$

$\therefore -x[x^2-1]$

$\therefore x+x^3$

NOW,

$$\int_{x=0}^{x=1} (x - x^3)\, dx$$

$$= \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1$$

$$= \frac{1}{4} - \frac{1}{4} \Rightarrow \frac{2}{8}$$

$$= \frac{1}{4}$$

Q.10] $\iint_R (x^2+y^2)\, dx\, dy$, by changing into polar form
R is circle $x^2+y^2=1$

soln:- The equation of circle is $x^2+y^2=1$, centre $(0,0)$ $r=1$
r varies from 0 to 1, θ varies from 0 to 2π

$$I = \int_0^{2\pi} \left[\int_0^1 (r^2\cos^2\theta + r^2\sin^2\theta) r\, dr\right] d\theta \quad \because x = r\cos\theta,\ y = r\sin\theta$$

$$= \int_0^{2\pi} \left(\int_0^1 r^3\, dr\right) d\theta$$

$$= \int_0^{2\pi} \frac{1}{4}[r^4]_0^1\, d\theta$$

$$= \int_0^{2\pi} \frac{1}{4}[1]\, d\theta$$

$$= \frac{1}{4}\int_0^{2\pi} d\theta$$

$$= \frac{1}{4}[\theta]_0^{2\pi}$$

$$= \frac{2\pi}{4}$$

$$I = \frac{\pi}{2}$$